

SAUMEN BISWAS

Senior / Lead SDET | AI-Assisted Testing, Automation Frameworks & Platform Reliability

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PROFESSIONAL SUMMARY

Senior Software Development Engineer in Test and technical leader with 15+ years of experience building scalable test automation, quality engineering practices, and reliability systems for complex, high-stakes platforms. Deep expertise in Playwright and Python, AI-assisted testing (LLM-orchestrated generation, intelligent selection, and self-healing), and driving measurable improvements in quality and delivery speed. Proven track record of reducing flaky failures by **73%** and improving CI/CD runtime by **60%** while mentoring engineers and partnering cross-functionally. Strong background in fintech and payments environments (Visa, Upstart) with a focus on production-quality code, framework ownership, and raising the engineering bar. Available immediately; authorized to work in the U.S. with H-1B portability.

CORE COMPETENCIES

- **Test Automation & Frameworks** – Designing and owning scalable automation (Playwright, pytest, API, UI, end-to-end)
- **AI-Assisted Quality Engineering** – LLM-driven test generation, intelligent prioritization, self-healing systems, and judgment-based curation
- **Platform Reliability & Observability** – Metrics, dashboards, failure triage, and continuous improvement of quality signal
- **CI/CD & Quality Gates** – Integrating automated tests into pipelines for fast, trustworthy feedback
- **Technical Leadership & Mentorship** – Raising team capability through code/test review, pairing, and standards definition
- **Domain Experience** – Large-scale fintech and global payments systems
- **Tools:** Python (strong), Playwright, pytest, TypeScript/JavaScript, Git, CI/CD platforms, observability tools, AI coding assistants (Claude, Cursor)

PROFESSIONAL EXPERIENCE

Senior Software Developer in Test → Technical Lead, Automation & AI-Assisted Quality

Upstart Network, Inc., San Mateo, CA | Apr 2022 – Present

- Designed, built, and evolved scalable test automation frameworks and AI-assisted quality systems used across 18 engineering teams.
- Developed LLM-orchestrated intelligent test selection and self-healing automation that reduced flaky failures by **73%** and improved CI/CD runtime by **60%**.

- Partnered early with product and engineering teams to improve testability, shape quality standards, and embed reliability into the development lifecycle.
- Owned quality metrics, dashboards, and failure triage processes that provided clear visibility into release risk and system health.
- Mentored engineers on automation patterns, AI-assisted workflows, and production-quality test code through review, pairing, and hands-on guidance.

Senior Software Engineer Teladoc Health, Mountain View, CA | Nov 2020 – Mar 2022

- Built and improved automation frameworks and quality practices for large-scale platforms.
- Collaborated with engineering teams on reliability, debugging, and continuous improvement of delivery processes.

Senior Staff Software Engineer Visa Inc., Foster City, CA | Dec 2014 – Oct 2020

- Led quality engineering and automation initiatives for high-scale global payment systems.
- Established engineering standards, scalable frameworks, and best practices that improved reliability and operational excellence.
- Mentored engineers and influenced technical direction across teams in a complex, regulated payments environment.
- Received the **Visa Go Beyond Innovation Award (2017, 2019)** for building scalable quality and automation systems with measurable impact.

Senior Software Engineer R S Software (India) Ltd., Kolkata | Mar 2007 – Nov 2014

- Designed and maintained software and test automation solutions for complex enterprise systems.

KEY IMPACT HIGHLIGHTS

- Reduced flaky test failures by **73%** and improved CI/CD runtime by **60%** through intelligent automation and framework improvements.
- Built and standardized automation frameworks adopted across multiple engineering teams.
- Applied AI/LLM techniques to test generation, prioritization, and self-healing with strong engineering judgment.
- Delivered technical leadership and mentorship that raised quality practices and engineering maturity.
- Operated effectively in high-stakes fintech and global payments environments requiring reliability and precision.

TECHNICAL EXPERTISE

- **Automation:** Playwright, pytest, API/UI/end-to-end frameworks, production-quality test code
- **AI-Assisted Testing:** LLM-orchestrated generation, intelligent selection, self-healing systems
- **Reliability:** Observability, metrics, failure analysis, quality gates, CI/CD integration
- **Leadership:** Mentoring, standards definition, cross-team influence, hands-on technical guidance
- **Domain:** Fintech platforms, global payments systems, high-reliability distributed environments
- **Tools:** Python (strong), TypeScript/JavaScript, Git, CI/CD platforms, observability tools, Claude, Cursor

EDUCATION & CERTIFICATIONS

- M.S. Information Technology Management, Western Governors University, 2020
- B.S. Electrical Engineering, WBUT, 2005
- CAPM (PMI), CSM (Scrum Alliance), CTFL (ASTQB), Machine Learning (Stanford Coursera)
- IEEE Senior Member (2025) | Fellow – BCS (2026)

AWARDS & RECOGNITION

- Visa Go Beyond Innovation Award (2017) – Scalable quality and automation systems for high-reliability payment platforms
- Publications on AI-enhanced testing methodologies
- Active contributor to quality engineering and AI testing communities